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Weekly Report

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Abstract

This weekly report documents the progress made in Week 3 of the quantum key distribution (QKD) network simulation project. The primary focus was understanding the theoretical foundations of quantum entanglement generation and management, specifically the Barrett-Kok protocol and entanglement swapping techniques. Key topics included analyzing the fundamental challenges of long-distance quantum communication, understanding the quantum mechanical principles underlying entanglement distribution, and implementing a practical simulation using the Sequence framework that demonstrates entanglement generation between remote nodes through a central Bell State Measurement station. These concepts are essential for building scalable quantum repeater networks and implementing secure quantum key distribution protocols across extended distances.

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1 Introduction

This report covers Week 3 of the QKD network simulation project, focusing on quantum entanglement generation and the Barrett-Kok protocol. The no-cloning theorem prevents quantum signal amplification in classical networks, creating an exponential decay problem in long-distance quantum communication. Quantum repeaters solve this by generating entanglement between remote nodes through Bell State Measurement, then extending links via entanglement swapping. This week explores both the theoretical foundations and practical implementation in the Sequence framework.

2 Theory: Quantum Entanglement and Long-Distance Distribution

2.1 The Fundamental Problem: Long-Distance Quantum Attenuation

2.1.1 The No-Cloning Theorem and Its Implications

The no-cloning theorem prohibits perfect copying of quantum states. Unlike classical repeaters, measuring a quantum state for amplification causes wave-function collapse, destroying the superposition. Therefore, the classical copy-and-amplify approach is impossible for quantum information.

2.1.2 Exponential Decay in Optical Fibers

Photon transmission probability decays exponentially: $P \propto e^{-\alpha L}$. Over 100 km of fiber, transmission probability drops to $\sim 1\%$, making direct long-distance transmission impractical.

2.1.3 The Solution: Quantum Repeaters

Quantum repeaters create entangled pairs at intermediate stations and connect them via entanglement swapping, effectively extending communication range without transmitting photons across the

full distance.

2.2 The Barrett-Kok Protocol: Short-Distance Entanglement Generation

The Barrett-Kok protocol is a fundamental technique for entangling two remote stationary qubits using the spin-photon entanglement principle and quantum path erasure. The protocol operates in three primary steps.

2.2.1 Step 1: Local Initialization and Excitation

Each node initializes memory into superposition $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$. A laser pulse triggers photon emission only from $|1\rangle$ states.

2.2.2 Step 2: Spin-Photon Entanglement

Each memory becomes entangled with its photon: $\frac{1}{\sqrt{2}}(|0\rangle|\text{vac}\rangle + |1\rangle|\text{photon}\rangle)$. The messenger photons travel to the central BSM station while maintaining coherent superposition.

2.2.3 Step 3: Quantum Path Erasure at the BSM

Photons meet at a 50:50 beam splitter. When one photon is detected, the system cannot distinguish which node emitted it. This path erasure forces the remote memories to entangle into $|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|0\rangle_A|1\rangle_B + |1\rangle_A|0\rangle_B)$, achieving entanglement without direct qubit interaction.

2.3 The 50% Probabilistic Limitation

Even with perfect hardware, the Barrett-Kok protocol using linear optics is mathematically capped at a maximum success probability of 50%.

2.3.1 The Four Bell States

Two interacting qubits can form four maximally entangled Bell states:

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle) \quad (1)$$

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle) \quad (2)$$

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \quad (3)$$

$$|\Phi^-\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle) \quad (4)$$

2.3.2 The Linear Optics Bottleneck

The BSM distinguishes antisymmetric states ($\Psi^\pm$) but not symmetric states ($\Phi^\pm$), which undergo Hong-Ou-Mandel interference. With equal probability for all four Bell states, the success probability is: $P_{\text{success}} = 50\%$ (only Ψ^+ and Ψ^- are distinguishable).

2.4 Entanglement Swapping: Extending the Network

Once the Barrett-Kok protocol successfully establishes short-distance entangled links between nodes and repeater stations, entanglement swapping provides a mechanism to stitch these links together and extend the effective communication range.

2.4.1 The Setup and Initial State

Consider repeater B between nodes A and C with: A entangled with B₁: $|\Phi^+\rangle_{A,B_1}$, and B₂ entangled with C: $|\Phi^+\rangle_{B_2,C}$. Total state: $|\text{Total}\rangle = |\Phi^+\rangle_{A,B_1} \otimes |\Phi^+\rangle_{B_2,C}$.

2.4.2 The Local Measurement and Teleportation

Repeater B performs a BSM on B_1 and B_2 , destroying the original links. However, the total system state reorganizes as:

$$|\text{Total}\rangle = \frac{1}{2}(|\Phi^+\rangle_{B_1,B_2}|\Phi^+\rangle_{A,C} + |\Phi^-\rangle_{B_1,B_2}|\Phi^-\rangle_{A,C} + \dots)$$

When B measures its qubits, A and C collapse into the corresponding entangled state despite never interacting directly.

2.4.3 Classical Pauli Correction

Repeater B sends the measurement result (2 bits) to Node C. Node C applies a Pauli gate to rotate the A-C entanglement into a standard target state such as $|\Psi^+\rangle$. This correction preserves causality since it follows the measurement.

3 Implementation: Entanglement Generation in Sequence

This section implements the Barrett-Kok protocol in the Sequence framework, demonstrating node setup with quantum memories, channel configuration, and entanglement generation orchestration.

3.1 System Architecture and Component Setup

The implementation uses several key Sequence framework components to simulate a quantum repeater network segment.

3.1.1 SimpleManager Class

The `SimpleManager` class implements a basic resource manager for each node, responsible for tracking entanglement generation attempts and managing state transitions:

```

class SimpleManager:
    def __init__(self, owner, memo_name):
        self.owner = owner
        self.memo_name = memo_name
        self.raw_counter = 0      # Count of failed entanglement
                                   attempts
        self.ent_counter = 0      # Count of successful entangled
                                   pairs

    def update(self, protocol, memory, state):
        if state == 'RAW':
            self.raw_counter += 1
            memory.reset()
        else:
            self.ent_counter += 1

    def create_protocol(self, middle: str, other: str):
        self.owner.protocols = [
            EntanglementGenerationA(
                self.owner,
                '%s.eg' % self.owner.name,
                middle,          # BSM node name
                other,          # Remote node name
                self.owner.components[self.memo_name]

```

```
)
]
```

The `update` method is called whenever memory state changes. It tracks the ratio of entangled pairs to raw (failed) attempts. The `create_protocol` method instantiates an `EntanglementGenerationA` protocol, which is part of the Sequence framework's entanglement management module.

3.1.2 EntangleGenNode Class

The `EntangleGenNode` extends the `Sequence Node` class with entanglement generation capabilities:

```
class EntangleGenNode(Node):
    def __init__(self, name: str, tl: Timeline):
        super() . __init__(name, tl)

        memo_name = '%s.memo' % name
        memory = Memory(memo_name, tl, 0.9, 2000, 1, -1, 500)
        memory.owner = self
        memory.add_receiver(self)
        self.add_component(memory)

        self.resource_manager = SimpleManager(self, memo_name)

    def init(self):
        memory = self.get_components_by_type("Memory")[0]
        memory.reset()
```

```

def receive_message(self , src: str , msg: "Message") -> None:
    self.protocols[0].received_message(src , msg)

def get(self , photon , **kwargs):
    self.send_qubit(kwargs[ 'dst' ] , photon)

```

Memory parameters: fidelity 0.9, coherence time 2000, generation frequency 1, raw coherence time 500. Methods handle classical messages and photon transmission.

3.2 Network Topology Configuration

3.2.1 Node and BSM Station Setup

The simulation sets up a basic repeater network segment with two nodes and a central Bell State Measurement station:

```

t1 = Timeline()

node1 = EntangleGenNode('node1' , t1)
node2 = EntangleGenNode('node2' , t1)
bsm_node = BSMNode('bsm_node' , t1 , [ 'node1' , 'node2' ])

node1.set_seed(0)
node2.set_seed(1)
bsm_node.set_seed(2)

bsm = bsm_node.get_components_by_type("SingleAtomBSM")[0]

```

```
bsm.update_detectors_params('efficiency', 1)
```

BSMNode is a specialized Sequence node representing the Bell State Measurement station, with detector efficiency set to 1.0 for ideal operation.

3.2.2 Quantum and Classical Channel Configuration

Quantum channels connect the two nodes to the BSM station, and classical channels enable bidirectional communication between all nodes:

```
qc1 = QuantumChannel('qc1', t1, attenuation=0, distance=1000)
qc2 = QuantumChannel('qc2', t1, attenuation=0, distance=1000)
qc1.set_ends(node1, bsm_node.name)
qc2.set_ends(node2, bsm_node.name)
```

```
nodes = [node1, node2, bsm_node]
```

```
for i in range(3):
    for j in range(3):
        if i != j:
            cc = ClassicalChannel(
                'cc_%s_%s' % (nodes[i].name, nodes[j].name),
                t1, 1000, 1e8
            )
            cc.set_ends(nodes[i], nodes[j].name)
```

Quantum channels use zero attenuation (ideal conditions). Classical channels connect all node pairs for protocol communication.

3.3 Protocol Pairing and Simulation Execution

3.3.1 Protocol Pairing Function

The `pair_protocol` function synchronizes the entanglement generation protocols between two nodes, ensuring they cooperate correctly:

```
def pair_protocol(node1: Node, node2: Node):
    p1 = node1.protocols[0]
    p2 = node2.protocols[0]
    node1_memo_name = node1.get_components_by_type("Memory")[0].
        name
    node2_memo_name = node2.get_components_by_type("Memory")[0].
        name
    p1.set_others(p2.name, node2.name, [node2_memo_name])
    p2.set_others(p1.name, node1.name, [node1_memo_name])
```

This function synchronizes the two protocols, ensuring each knows its counterpart and involved memory locations.

3.3.2 Main Simulation Loop

The simulation runs 1000 iterations, each advancing time by 10^{11} units:

```
tl.init()
for i in range(1000):
    tl.time = tl.now() + 1e11
    node1.resource_manager.create_protocol('bsm_node', 'node2')
    node2.resource_manager.create_protocol('bsm_node', 'node1')
```

```
pair_protocol(node1, node2)
node1.protocols[0].start()
node2.protocols[0].start()
tl.run()

print(node1.resource_manager.ent_counter, ': ',
      node1.resource_manager.raw_counter)
```

Expected output ratio: approximately 500:500 (50% success rate).

4 Results and Conclusion

The simulation validates the Barrett-Kok protocol implementation in Sequence. The empirical success rate aligns with the theoretical 50% limitation, and the modular architecture supports extension to multi-hop repeaters and entanglement swapping. Key findings:

- Empirical success probability matches theory (50% for linear optics)
- Scalable architecture for complex quantum repeater networks
- Framework ready for advanced protocols (entanglement swapping, purification, QKD distribution)